Theory Concepts & Definitions

PR. 2 — Data Cleanser: Patient Health Records

Missing Value Imputation & Outlier Treatment

1. Missing Data

Missing data occurs when no value is recorded for a variable in an observation. In healthcare records, this commonly arises from inconsistent reporting, skipped measurements, or transcription errors.

Types of Missingness

Type	Definition
MCAR — Missing Completely at Random	The probability of missingness is unrelated to any observed or unobserved data.
MAR — Missing at Random	The probability of missingness depends on observed data, but not the missing value itself.
MNAR — Missing Not at Random	The probability of missingness depends on the missing value itself (e.g., very sick patients skip a test).

2. Imputation Techniques

Simple Imputer (Numerical)

Replaces missing numeric values with a single summary statistic — typically the mean or median of the observed values in that column. Mean imputation is simple and unbiased under MCAR, but understates variance and distorts correlations between features. Median imputation is more robust to skew and outliers than mean.

Simple Imputer (Categorical) / Most Frequent Imputation

Replaces missing categorical values with the mode (most frequently occurring category). Appropriate for low-cardinality categorical fields with no strong dependency on other variables.

Missing Indicator + Random Sample Imputation

Two complementary ideas combined: (1) a binary 'was missing' indicator column preserves the information that a value was originally absent — which can itself be predictive; (2) random sample imputation fills gaps by randomly drawing from the column's own observed distribution, preserving its original shape better than a single constant value would.

KNN Imputer

A multivariate technique that fills a missing value using the average (or distance-weighted average) of the k most similar rows (nearest neighbors), based on the other available features. This captures relationships between variables that simple imputation ignores, at the cost of higher computational complexity.

MICE — Multiple Imputation by Chained Equations

An iterative, multivariate technique that models each column with missing values as a function of all other columns, cycling through columns repeatedly until the imputed values stabilize. It generally preserves inter-variable relationships and distributional shape better than single-pass methods, making it well suited to datasets — like this one — where features are physiologically correlated (age, BMI, blood pressure).

3. Outliers

An outlier is a data point that differs significantly from other observations. In clinical data, outliers can represent genuine (if rare) extreme measurements, or they can result from equipment error, data-entry mistakes, or unit-conversion issues — distinguishing between the two is a core part of the analyst's judgment.

Detection Methods

Method	Definition	Sensitivity
Z-score method	Flags points more than a chosen number of standard deviations (commonly 3) from the mean.	Assumes roughly normal data; only catches extreme-tail values.
IQR method	Flags points below Q1 − 1.5×IQR or above Q3 + 1.5×IQR, where IQR = Q3 − Q1.	Distribution-free; more sensitive to moderate skew.
Percentile method	Flags/caps points below a low percentile (e.g., 1st) or above a high percentile (e.g., 99th).	Directly controls what fraction of data is treated as extreme.

Treatment Methods

Method	Definition	Trade-off
Removal	Deletes rows identified as outliers.	Simple, but discards potentially valid data and shrinks sample size.
Winsorization	Caps (clips) extreme values to a chosen percentile bound rather than deleting them.	Retains every row; reduces the influence of extremes on mean/variance without losing patients.

In this project, Winsorization was preferred to outright removal specifically because the dataset is clinical: an extreme glucose or cholesterol reading is often a genuinely high-risk patient rather than a data-entry error, and the downstream model's purpose is to identify exactly those patients.

4. Why Data Cleaning Matters for Machine Learning

- Most ML algorithms cannot handle missing values directly and will error out or silently drop rows, biasing the training sample.
- Unaddressed outliers can dominate the loss function of distance-based or linear models, skewing coefficients and predictions.
- Consistent, well-imputed, outlier-controlled data improves model stability, generalization, and interpretability.
- A documented, comparison-driven cleaning process (rather than a single default choice) makes the resulting model's behavior easier to audit and trust — especially important in healthcare applications.

Summary

Missing-value imputation and outlier treatment are not mechanical checkbox steps — each technique makes a different implicit assumption about the data-generating process. Comparing multiple methods side-by-side, as done in this project, turns an otherwise arbitrary choice into an evidence-based one, and is a hallmark of mature, production-grade data preparation.